

VERKHNETULOMSKIY, Russian Federation

WMO: 221060

Lat: 68.6083N

Lon: 31.8039E

Elev: 63

StdP: 100.57

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.6	-30.2	-36.3	0.1	-33.3	-32.7	0.2	-30.1	5.6	-4.5	4.9	-5.1	0.1	160	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	24.7	17.1	22.3	15.9	20.2	14.8	18.2	23.1	16.7	21.1	15.4	19.5	2.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	11.5	21.1	14.6	10.5	19.3	13.3	9.6	17.5	51.5	23.5	47.0	21.2	43.3	19.5	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.8	4.2	3.5	-36.2	27.8	3.7	1.9	-38.8	29.1	-41.0	30.3	-43.1	31.3	-45.8	32.7		
			DB	WB	-36.1	19.6	3.6	1.5	-38.6	20.7	-40.7	21.6	-42.7	22.4	-45.3	23.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.2	-12.9	-12.6	-6.4	-0.3	4.8	10.1	13.9	11.7	7.3	1.0	-5.8	-9.3
	DBStd	10.69	8.51	8.26	5.84	4.28	4.11	4.17	3.99	3.51	3.54	4.43	6.35	7.43
	HDD10.0	3861	711	632	509	310	171	49	10	21	97	281	473	598
	HDD18.3	6636	969	865	768	560	419	250	146	209	332	538	723	856
	CDD10.0	278	0	0	0	0	9	51	131	72	14	0	0	0
	CDD18.3	13	0	0	0	0	0	2	9	3	0	0	0	0
	CDH23.3	127	0	0	0	0	2	17	88	20	0	0	0	0
	CDH26.7	18	0	0	0	0	1	2	14	2	0	0	0	0
Wind	WSAvg	1.5	1.2	1.3	1.6	1.6	1.8	1.9	1.6	1.3	1.5	1.5	1.3	1.3
Precipitation	PrecAvg	537	34	27	29	30	40	62	74	64	51	54	35	37
	PrecMax	661	63	48	60	80	60	124	135	144	101	119	68	67
	PrecMin	409	16	10	8	11	16	10	25	21	12	14	9	16
	PrecStd	72	11	10	11	14	14	31	27	30	23	26	13	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.5	3.3	5.5	12.6	21.5	25.9	28.7	25.9	18.4	11.5	6.0	3.7
		MCWB	1.0	1.7	2.3	6.9	13.4	17.7	18.9	18.1	13.9	8.9	4.9	2.1
	2%	DB	0.5	1.5	3.5	8.8	17.3	22.3	25.7	22.6	16.0	9.7	3.7	2.5
		MCWB	-0.4	0.2	1.1	4.7	10.8	15.0	17.6	16.7	12.6	7.7	2.7	1.2
	5%	DB	-1.0	0.1	1.9	7.0	14.6	20.1	23.5	19.9	14.2	8.2	2.5	1.0
		MCWB	-1.8	-1.1	0.0	3.5	9.4	13.7	16.5	15.2	11.6	6.7	1.6	0.1
	10%	DB	-2.7	-1.5	0.8	5.4	12.2	18.0	21.4	17.7	12.5	6.8	1.4	-0.5
		MCWB	-3.3	-2.5	-0.8	2.6	8.0	12.6	15.7	14.3	10.4	5.6	0.6	-1.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	2.0	2.6	7.2	14.6	19.2	20.2	18.9	14.6	9.2	5.1	2.2
		MCDB	2.2	3.1	4.9	12.2	20.4	24.6	25.5	24.4	17.6	11.0	6.0	3.3
	2%	WB	-0.3	0.3	1.3	5.0	11.5	15.9	18.7	17.3	12.9	8.0	2.9	1.3
		MCDB	0.4	1.4	3.0	8.2	16.5	20.9	24.0	21.4	15.3	9.6	3.5	2.4
	5%	WB	-1.7	-1.1	0.3	3.9	9.7	14.2	17.4	15.8	11.7	6.7	1.7	0.1
		MCDB	-1.1	0.1	1.6	6.5	14.0	19.0	22.2	19.3	13.9	8.0	2.4	1.0
	10%	WB	-3.2	-2.5	-0.7	2.8	8.2	12.9	16.1	14.4	10.6	5.6	0.7	-1.3
		MCDB	-2.6	-1.5	0.7	5.0	12.0	17.6	20.7	17.4	12.4	6.7	1.3	-0.5
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.8	8.4	7.7	7.3	8.0	8.4	7.6	6.3	4.2	5.2	6.3
		MCDBR	6.0	5.0	6.7	9.4	12.7	12.1	13.0	12.0	9.1	4.8	3.7	4.6
		MCWBR	5.5	4.3	5.1	5.9	7.4	6.2	6.1	6.4	5.8	3.7	3.5	4.2
	5% WB	MCDBR	5.7	5.0	5.9	8.5	11.7	10.7	10.8	10.0	8.3	4.4	3.6	4.5
		MCWBR	5.4	4.4	4.9	5.5	7.1	6.1	6.0	6.1	5.5	3.5	3.4	4.2
Clear-Sky Solar Irradiance	taub	0.206	0.257	0.302	0.335	0.358	0.355	0.372	0.354	0.319	0.278	0.360	N/A	
	taud	1.913	2.049	2.100	2.155	2.336	2.431	2.412	2.437	2.503	2.244	2.669	N/A	
	Ebn at Noon	55	501	704	797	828	844	813	786	724	486	94	0	
	Edh at Noon	7	47	84	108	102	96	95	83	59	39	10	0	
All-Sky Solar Radiation	RadAvg	0.03	0.43	1.81	3.44	4.38	4.79	4.40	3.03	1.66	0.50	0.07	0.00	
	RadStd	0.00	0.05	0.11	0.22	0.46	0.43	0.54	0.32	0.16	0.07	0.01	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page